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Task Team Lead: Agustín Samano

Region: Malaysia (pilot) → Southeast Asia (extension)

Timeline: US Benchmark January 2026; Malaysia Pilot February 2026; Multi-Country Dataset June 2026

1 Motivation

Fiscal policy is one of governments’ most powerful levers for macroeconomic stability and private-sector development. Yet for most emerging markets we lack the cornerstone input required for credible analysis: **consistent and comparable measures of exogenous fiscal shocks**¹.

The United States is the only country where this has been done systematically, through the “narrative approach” pioneered by Romer and Romer (2010). Their method uses historical documents—budget speeches, economic reports, legislative records—to identify why taxes or spending changed, and to distinguish exogenous shocks from policy actions responding to the business cycle.

Replicating this approach manually is costly. For low- and middle-income countries, it has simply never been feasible.

Recent advances in Large Language Models (LLMs) make this possible for the first time. These models can read long historical documents, interpret motivations, and extract structured information at scale—opening the door to producing robust fiscal shock series for developing countries.

This project proposes to build a **validated LLM-assisted methodology** for fiscal shock identification, starting with Malaysia and extending across Southeast Asia.

2 Novelty and Innovation

2.1 A. A US-Validated Codebook System

The project develops 4 domain-specific codebooks (C1-C4) validated on 44 US fiscal acts with Romer & Romer’s original labels using the Halterman & Keith (2025) validation framework. This ensures:

- we anchor the method in a gold-standard dataset,
- we quantify codebook accuracy against known results via rigorous behavioral testing,
- we design country-agnostic definitions that transfer across countries.

¹ See for example Mertens and Ravn (2013) and Romer and Romer (2010)

This step is essential for **credibility**: codebooks must demonstrate strong performance on US data before cross-country deployment.

2.2 B. Transfer Learning with Expert Validation

We deploy the US-validated codebooks to Malaysia, where English-language archives are extensive. The pipeline:

1. identifies fiscal measures meeting the “significant mention” rule (Codebook C1),
2. classifies motivations using Romer & Romer’s categories (Codebook C2),
3. extracts implementation timing using the midpoint rule (Codebook C3),
4. extracts fiscal impact magnitude (Codebook C4), and
5. integrates systematic expert validation following Halterman & Keith error analysis methodology.

Once methodology is validated on Malaysia (estimated 20-40 acts, 1980-2022), the pipeline extends to Indonesia, Thailand, the Philippines, and Vietnam, with built-in multilingual capability.

3 Project Outputs (June 2026)

3.1 A. New Data Assets

- Narrative fiscal episode datasets for 5 Southeast Asian economies (estimated 20-60 acts per country depending on political stability and archive quality).
- Expert-validated shock-event datasets identifying exogenous tax and spending changes: timing, size, and motivation.
- Harmonized multi-country panel suitable for macro and micro analysis.

3.2 B. Analytical Products

- Macro impulse responses estimated with local projections.
- Firm-level effects (investment, employment) using modern LP-DiD methods (Dube et al., n.d.).
- A methodological paper on LLM-assisted narrative identification, similar to (Romer and Romer, n.d.).

3.3 C. Tooling and Capacity

- A fully documented, open-source pipeline (R-based).
- Country-agnostic LLM codebooks, rules, and evaluation protocols.
- A tested architecture that can be extended to Africa, South Asia, and Latin America.

4 Implementation Plan

Phase 0 – Codebook Development (Jan 2026)

Develop and validate 4 codebooks (C1-C4) on 44 US fiscal acts using the Halterman & Keith framework: codebook preparation, behavioral testing, zero-shot evaluation, and error analysis. Validate performance against Romer & Romer ground truth.

Phase 1 – US Full Production (Jan-Feb 2026)

Run validated codebooks on the full US document corpus. Compare end-to-end results against Romer & Romer’s published shock series to confirm production readiness.

Phase 2 – Malaysia Pilot (Feb-Mar 2026)

Deploy US-validated codebooks to Malaysia documents (1980-2022), generating candidate fiscal shock dataset. Engage local expert to validate outputs and assess cross-country transfer performance. Refine codebook definitions based on expert feedback.

Phase 3 – Southeast Asia Scaling (June 2026)

Extend to Indonesia, Thailand, the Philippines, and Vietnam, incorporating expert review and news-based cross-checks.

5 Risk Management

Risk	Mitigation
Limited training data (44 US acts)	Few-shot learning; expert validation loop; focus on methodology over scale.
Weak or uneven archives	Combine parliamentary, budget, and press sources; clearly document gaps.
Cross-country transfer failure	Expert validation quantifies performance; error analysis informs refinement.
LLM misclassification	US benchmarking + systematic expert review; conservative thresholds for exogeneity.
Translation quality for non-English sources	Native multilingual LLMs (Claude/GPT-4); manual review of samples.
Political bias in source documents	Independent cross-validation with contemporaneous news archives.

6 Strategic Value for the World Bank

1. Transforms a previously infeasible task into a reproducible method for potentially dozens of client countries.
2. Provides a missing input for macroeconomic modeling enabling better fiscal forecasting and policy advice.
3. Offers the first systematic estimates of firm-level impacts of fiscal policy in Southeast Asia using modern causal methods.
4. Establishes the World Bank as a pioneer in responsible, auditable LLM use for economic analysis.
5. Creates a transferable framework that can serve as a global public good for future fiscal reform programs.

Conclusion

This project brings together frontier methodology, practical policy relevance, and realistic scalability. The US-benchmarked codebook system (C1-C4) demonstrates the Halterman & Keith validation framework applied to economic history; the Malaysia pilot tests cross-country transfer with expert validation; and the multi-country rollout provides methodology replicable beyond Southeast Asia. The contribution is methodological: showing country-agnostic codebooks can assist experts in fiscal shock identification, reducing effort from months to weeks, even with limited training data.

References

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